Cones over the Paw and the Triangle–Edge Product

Abstract

Let P be the paw graph, and let $R = K_3 \sqcup K_2$. For either $B = P \sqcup \overline{K_s}$ or $B = R \sqcup \overline{K_s}$, we prove that the cone $H = K_1 \vee B$ satisfies

$$t(H, W) \geq \Phi_H(p)$$

for every graphon of edge density $p \geq 2/3$. This is the complete required interval because $\chi(H) = 4$. The proof uses only the previously proved paw and triangle bounds, multiplicativity, and a fully proved weighted link argument. It is not a general universal-vertex closure. The result proves the previously open Atlas graphs 49, 156, and 165.

1 The two base families and their common target

Let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon on a probability space (Ω, μ). For a finite graph G, write

$$t(G, W) = \int_{\Omega^{V(G)}} \prod_{uv \in E(G)} W(x_u, x_v) \prod_{u \in V(G)} \mathrm{d}\mu(x_u).$$

Put

$$p = t(K_2, W), \qquad d(x) = \int_\Omega W(x, y)\, \mathrm{d}\mu(y).$$

The paw P is a triangle with one pendant edge. Let

$$R = K_3 \sqcup K_2.$$

For $s \geq 0$, define

$$B_s^P = P \sqcup \overline{K_s}, \qquad B_s^R = R \sqcup \overline{K_s}.$$

The isolated vertices have no effect on homomorphism densities.

The chromatic polynomials are

$$\chi_{B_s^P}(x) = x^{s+1}(x-1)^2(x-2), \qquad v(B_s^P) = s+4,$$
$$\chi_{B_s^R}(x) = x^{s+2}(x-1)^2(x-2), \qquad v(B_s^R) = s+5.$$

Consequently, both base families have the same target

$$\Phi_{B_s^P}(z) = \Phi_{B_s^R}(z) = \phi(z) := z^2(2z-1). \tag{1}$$

We use the following accepted positive inputs.

Theorem 1.1 (Base bounds)**.** *For every graphon V of edge density $z \geq 1/2$,*

$$t(B_s^P, V) \geq \phi(z), \qquad t(B_s^R, V) \geq \phi(z).$$

Proof. The paw is the member L_1 of the rooted triangle–tree theorem, so

$$t(P, V) \geq z^2(2z - 1).$$

Adding isolated vertices changes neither side. For the other family, multiplicativity over disjoint unions and the triangle Goodman bound give

$$t(R, V) = t(K_3, V)t(K_2, V) \geq z(2z - 1)z = z^2(2z - 1).$$

Again, isolates make no change. □

The polynomial ϕ satisfies

$$\phi(1/2) = 0, \qquad \phi'(z) = 2z(3z - 1) > 0, \qquad \phi''(z) = 12z - 2 > 0$$

on $[1/2, 1]$. It is therefore nonnegative, increasing, and convex on that interval.

Lemma 1.2 (Global tangent bound for the bases)**.** *Let B be any B_s^P or B_s^R, and fix $c \in [1/2, 1]$. Every graphon V, of arbitrary edge density $z \in [0, 1]$, satisfies*

$$t(B, V) \geq \phi(c) + \phi'(c)(z - c).$$

Proof. If $z \geq 1/2$, Theorem 1.1 and convexity give the result. If $z < 1/2$, the tangent line at c is at most $\phi(1/2) = 0$ when evaluated at $1/2$, and it decreases as its argument moves below $1/2$. Thus the tangent is nonpositive at z, while $t(B, V) \geq 0$. □

2 The weighted link estimate

Define

$$\tau(x) = \int_{\Omega^2} W(x, y)W(x, z)W(y, z) \, \mathrm{d}\mu(y) \, \mathrm{d}\mu(z).$$

Lemma 2.1 (Weighted rooted-triangle inequality)**.** *If $p > 0$, then for every integer $h \geq 0$,*

$$\int_\Omega d(x)^h \tau(x) \, \mathrm{d}\mu(x) \geq \frac{2p - 1}{p} \int_\Omega d(x)^{h+2} \, \mathrm{d}\mu(x).$$

Proof. Put $M_j = \int d^j$, let T_W be the graphon operator, and define

$$I_h = \int d^h \tau, \qquad J_h = \left\langle d^h, T_W d \right\rangle.$$

The inequality $ab \geq a + b - 1$ on $[0, 1]^2$, applied to $W(x, y), W(x, z)$ and multiplied by $d(x)^h W(y, z)$, gives

$$I_h \geq 2J_h - pM_h. \tag{2}$$

Set $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Then

$$T_W d = d - q\mathbf{1} + T_U u,$$

and hence

$$J_h = M_{h+1} - qM_h + \left\langle d^h, T_U u \right\rangle. \tag{3}$$

By symmetry of U, twice

$$\Big\langle d^h, T_U u \Big\rangle - \int d^h u^2$$

equals

$$\int_{\Omega^2} U(x,y)\big(u(y)-u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y),$$

which is nonnegative because $d = 1-u$. Thus

$$J_h \geq M_{h+2} - M_{h+1} + pM_h.$$

Substitution into (2) yields

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

Finally, subtracting $(2p-1)M_{h+2}/p$ from the right-hand side gives

$$\frac{1}{p}\int_\Omega d(x)^h (d(x)-p)^2\,\mathrm{d}\mu(x) \geq 0.$$

□

For $d(x) > 0$, set

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y),$$

and let W_x be the kernel W on (Ω, μ_x). Its edge density is $z_x = \tau(x)/d(x)^2$. At $d(x) = 0$, set $z_x = 0$.

Corollary 2.2 (Average density in the links)**.** *For every integer $N \geq 2$,*

$$\int_\Omega d(x)^N z_x\,\mathrm{d}\mu(x) \geq \left(2 - \frac{1}{p}\right)\int_\Omega d(x)^N\,\mathrm{d}\mu(x).$$

Proof. Apply Lemma 2.1 with $h = N-2$. □

3 The cone theorem

Theorem 3.1 (Special cones)**.** *Let B be any B_s^P or B_s^R, put $N = v(B)$, and define*

$$H = K_1 \vee B.$$

Every graphon W of edge density $p \geq 2/3$ satisfies

$$\boxed{t(H,W) \geq \Phi_H(p).}$$

Moreover, the exact target is

$$\Phi_H(p) = p^{N-3}(2p-1)^2(3p-2). \tag{4}$$

Since $\chi(H) = 4$, this is the complete required interval.

Proof. Put

$$c = 2 - \frac{1}{p}.$$

The hypothesis $p \geq 2/3$ gives $c \in [1/2, 1]$. Conditioning on the image x of the cone vertex gives the exact identity

$$t(H, W) = \int_\Omega d(x)^N t(B, W_x)\, \mathrm{d}\mu(x).$$

Let $M_N = \int d^N$. Lemma 1.2 gives

$$\begin{aligned} t(H, W) &\geq M_N \phi(c) \\ &\quad + \phi'(c) \left(\int d(x)^N z_x \, \mathrm{d}\mu(x) - c M_N \right). \end{aligned}$$

The second line is nonnegative by Corollary 2.2, since $c = 2 - 1/p$ and $\phi'(c) \geq 0$. Therefore

$$t(H, W) \geq M_N \phi(c) \geq p^N \phi(c),$$

where the last step is Jensen and $\phi(c) \geq 0$.

Now

$$c = \frac{2p-1}{p}, \qquad 2c - 1 = \frac{3p-2}{p}.$$

Consequently,

$$p^N \phi(c) = p^N c^2 (2c-1) = p^{N-3}(2p-1)^2(3p-2).$$

It remains to verify that this is the chromatic target. For a cone,

$$\chi_H(x) = x \chi_B(x-1).$$

If $q = 1 - p$, then $1/q - 1 = p/q$. Using (1) in its defining chromatic-polynomial form,

$$\begin{aligned} \Phi_H(p) &= q^{N+1} \frac{1}{q} \chi_B\left(\frac{p}{q}\right) \\ &= p^N \Phi_B\left(2 - \frac{1}{p}\right) = p^N \phi(c). \end{aligned}$$

This proves both the inequality and (4). □

4 Endpoints, equality, and catalogue consequences

At $p = 2/3$, the factor $3p - 2$ vanishes, so the target is zero. The balanced complete tripartite graphon has no K_4 and attains equality, because every graph in the theorem contains a K_4. At $p = 1$, both sides equal 1. More generally, every balanced complete k-partite graphon at $p = 1 - 1/k$, $k \geq 3$, attains equality by the chromatic polynomial identity.

Corollary 4.1 (New positive Atlas cases)**.** *The following previously open graphs are positive on the full interval $p \geq 2/3$:*

Atlas ID	*graph6*	*base after deleting cone vertex*	*target*
49	`Df{`	P	$p(2p-1)^2(3p-2)$
156	`E~H_`	$P \sqcup K_1$	$p^2(2p-1)^2(3p-2)$
165	`E~aG`	$K_3 \sqcup K_2$	$p^2(2p-1)^2(3p-2)$

5 Scope and hidden assumptions

The proof uses the rooted triangle–tree theorem for the paw, the triangle Goodman bound, and multiplicativity over disjoint unions. Every other analytic step is proved here.

This is not a general universal-vertex theorem. It applies only because the two stated base families share the explicitly verified target $\phi(z) = z^2(2z-1)$ and the required graphon lower bound. No conclusion is asserted for a different base graph.

The link tangent is valid even when an individual link has density below $1/2$: its tangent lower bound is then nonpositive. Thus the proof does not silently assume pointwise link density, regularity, minimum degree, injective copies, or finite-step graphons. It applies directly to every graphon and ordinary homomorphism densities.

A What the Lean formalization actually proves

Theorem 3.1 is formalized for both base families; Atlas 49, 156 and 165 are `verified`. The development is `lean/Taeyoung/Methods/PawCone/`, over the abstract `Methods/ConeBound.lean`. Two differences of substance.

Theorem 1.1 is proved, not accepted

Both base bounds are formalized rather than taken as inputs: the paw is L_1 of the rooted triangle–tree development, and $t(R,V) = t(K_3,V)t(K_2,V)$ uses Goodman, both of which this project proves. So no accepted graph-density result enters.

Lemma 1.2 is proved without convexity

$\phi'' \geq 0$ is never verified, and no convexity theory is used. The tangent bound is instead the polynomial identity

$$\phi(w) - \phi(c) - \phi'(c)(w-c) = (w-c)^2\big(2w + 4c - 1\big),$$

whose right-hand side is nonnegative as soon as $w, c \geq 1/2$; the case $z < 1/2$ is unchanged. Since both families share the target (1), everything after the base bound is proved once against an abstract base graph, and the two families are two instantiations of it.